



**NANYANG
TECHNOLOGICAL
UNIVERSITY**

CE4055

LAB:3

**Stealing Secrets from
Embedded Devices through
Side-Channel Attacks**



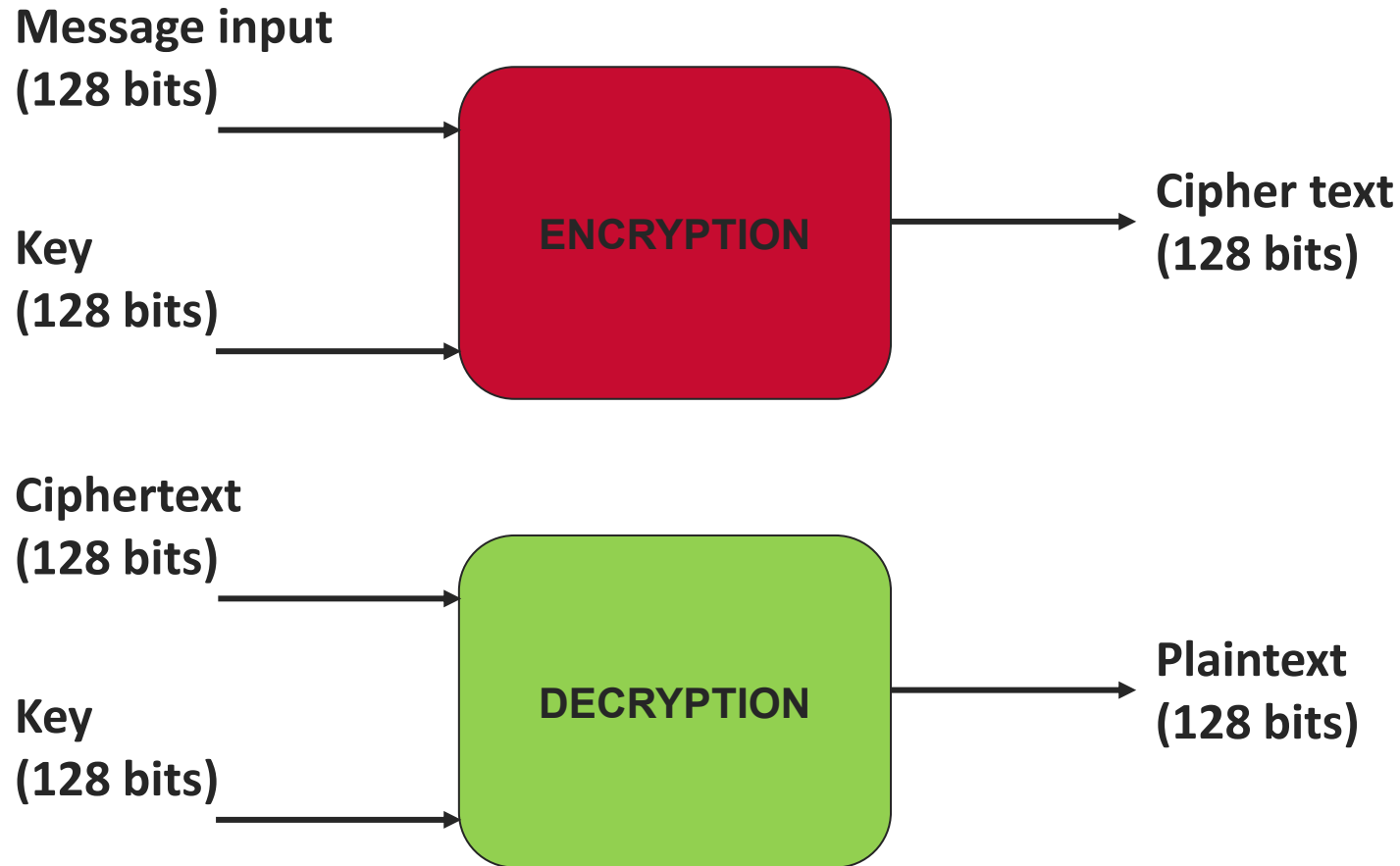
Acknowledgement (HW/SW Development)

- Arpan Jati
- Naina Gupta
- Ritu Ranjan Srivastava

AGENDA:

- **Understand Context of Side-Channel Attacks**
 - Why is it a Game changer in the field of cyber/hardware security?
- **Setup to perform Side-Channel Attack**
- **To inspire and instigate your interests in this topic!!**

BLOCK CIPHER: Used for Encryption/Decryption



- ❑ AES (Advanced Encryption Standard) is the most utilized encryption algorithm till date.

Is our crypto really fool proof???

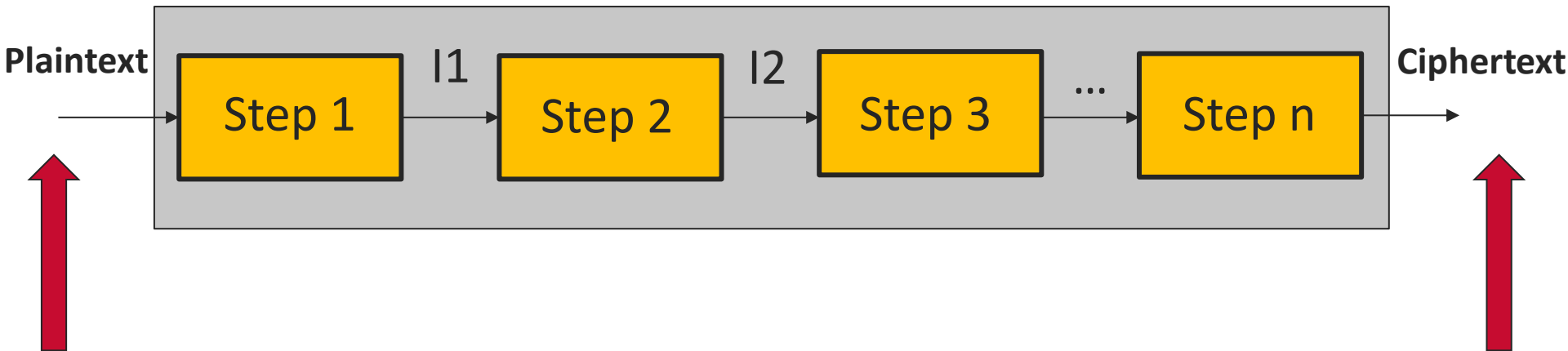
- ❑ To recover an AES-128 key: 70,000,000,000,000,000,000,000,000 years
- ❑ Even if you are given plaintexts and ciphertexts:
 - ❑ (pt_1, ct_1)
 - ❑ (pt_2, ct_2)
 - ❑ (pt_3, ct_3) and so on...
- ❑ It is computationally infeasible to recover the AES key...
- ❑ Not the end of the story!!!

Is our crypto really fool proof???

- ❑ **Caveat: Security proofs are only valid under a set of assumptions!!!**

Is our crypto really fool proof???

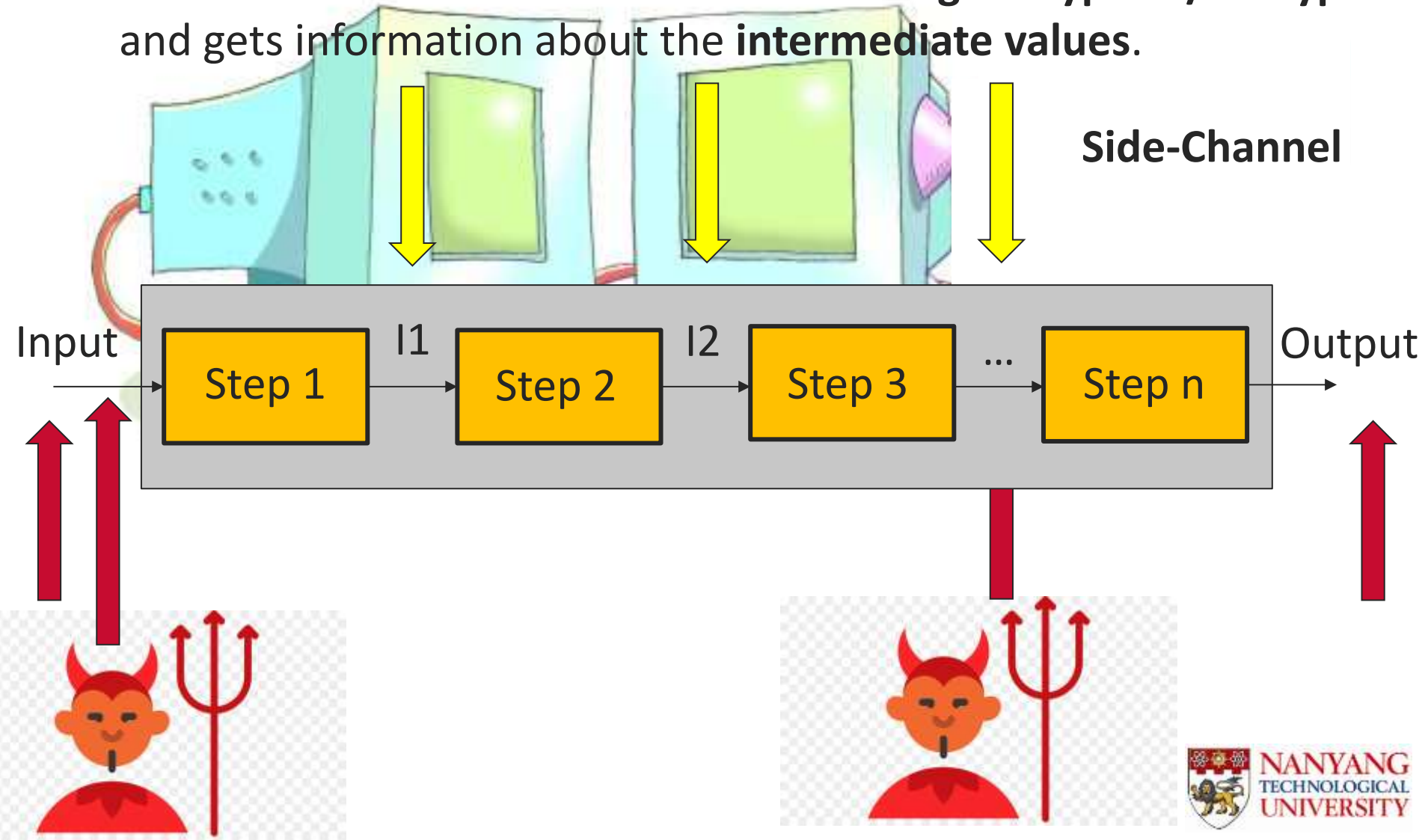
- ❑ **Blackbox Assumption:** Attacker only has access to the **input and output** (i.e) **plaintext and ciphertext**.



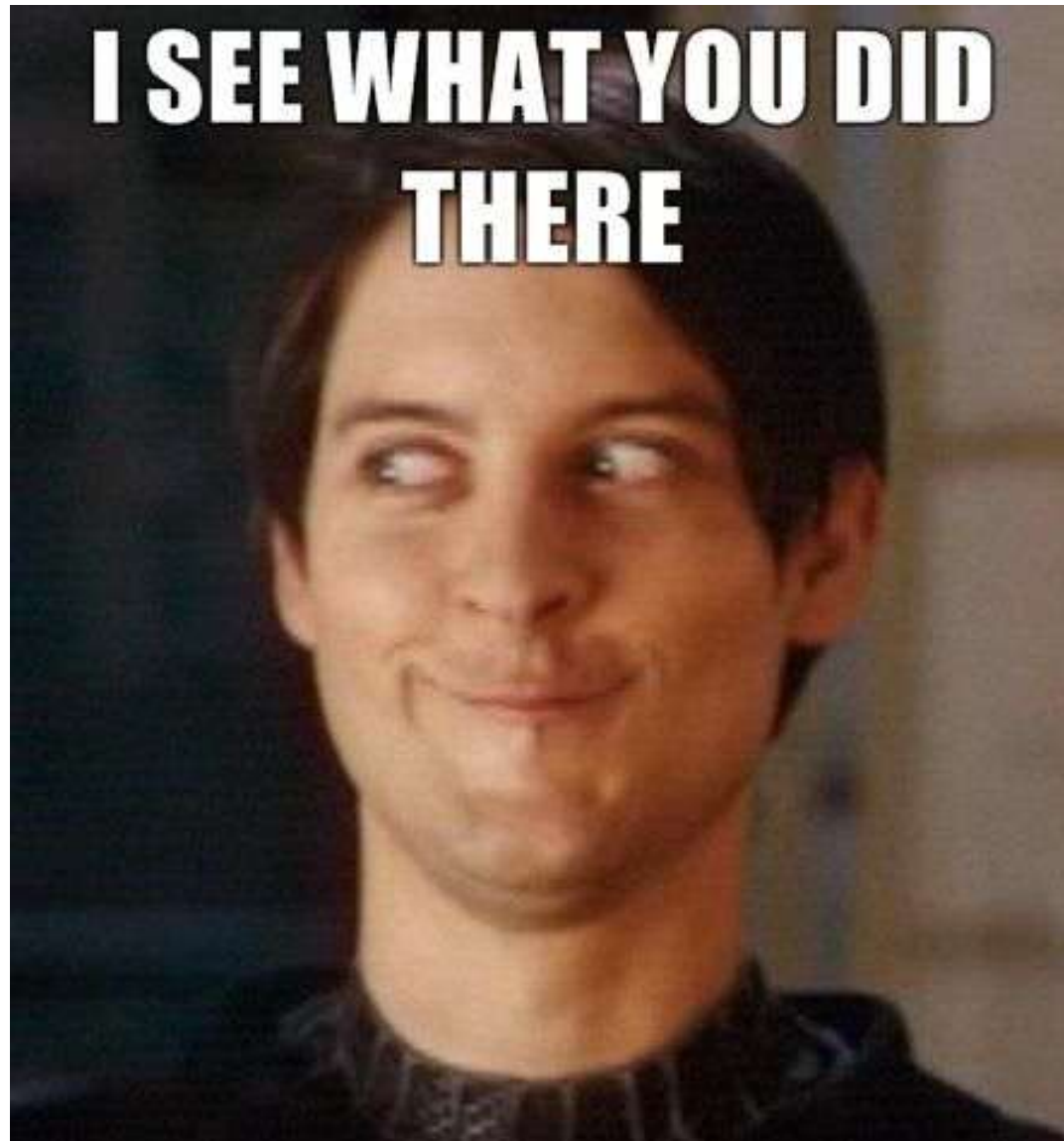
- ❑ **KEY QUESTION:** Is the black-box assumption true in practice?
- ❑ Can the attacker know more than just inputs and outputs??
- ❑ How can an attacker get information about intermediate data?

Is our crypto really fool proof???

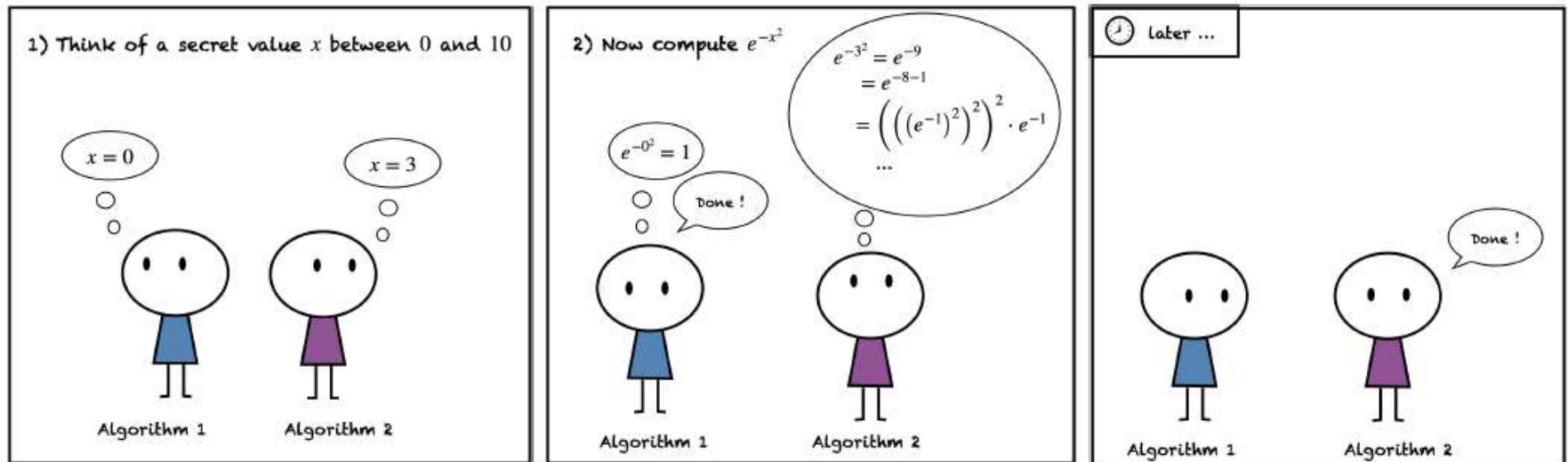
- ❑ Monitors the **behavior** of the device during encryption/decryption and gets information about the **intermediate values**.



Is our crypto really fool proof???

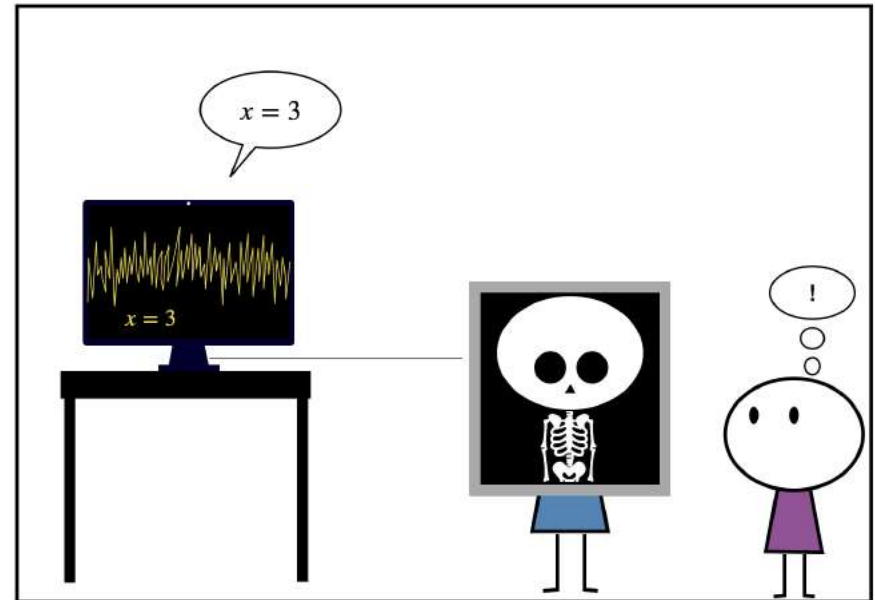
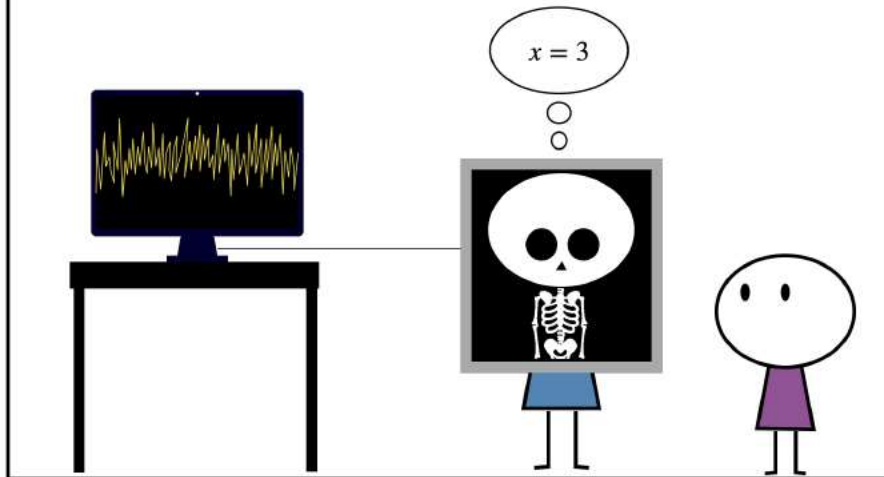


Timing Attack:



Power Analysis Attack:

Think of a secret value x between -10 and 10 while the X-ray machine records the emanations of your circuits



Side Channel Attacks:

- ❑ A cheating technique to perform cryptanalysis.
- ❑ There exist many attack vectors.

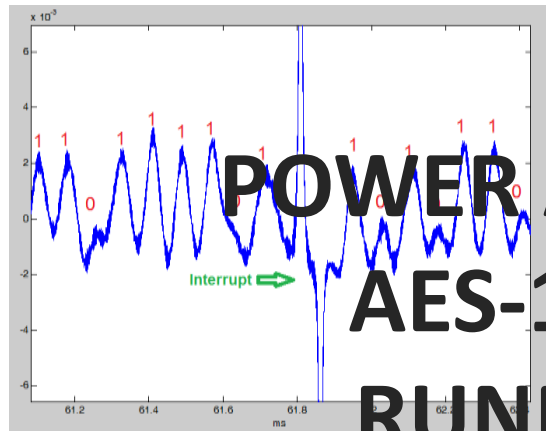


Fig. 1. Experimental setup for performing acoustic attacks.

**POWER ANALYSIS ATTACK ON
AES-128 BLOCK CIPHER
RUNNING ON AN 8-BIT
MICROCONTROLLER**

MELTDOWN
inside™

SPECTRE
inside™

intel

The image shows a blue background with the Intel logo in the center. Below the logo are two circular icons: one with a shield and the text 'MELTDOWN inside™' and another with a ghost and the text 'SPECTRE inside™'.

How do devices leak “info” through side-channels?

- ❑ Like humans (who belong to the same “homo sapiens” species), most electronic devices we use today also belong to the same species “CMOS technology”.
- ❑ So, most devices behave in a very similar manner they are performing computations (similar to micro-expressions) .

DONE WITH THE THEORY !!!

Target to Attack (Victim):

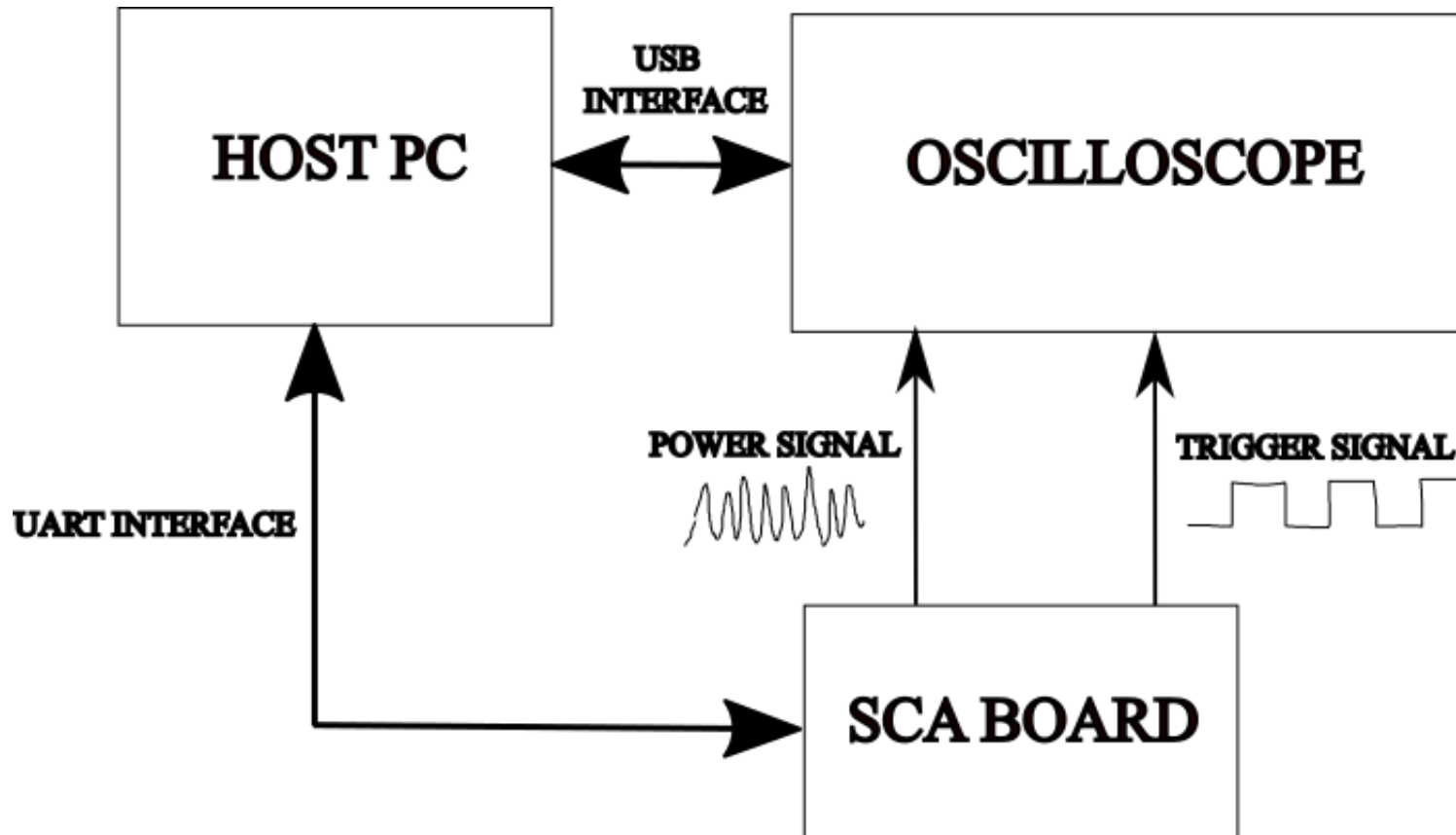
- ❑ **Device:** Atmega328P microcontroller running AES encryption
- ❑ **Attacker's Motivation:** To recover key used for encryption
- ❑ **Features:**
 - ❑ You can input plaintext to encrypt
 - ❑ It will generate ciphertext using secret key
 - ❑ Communication through UART interface
- ❑ Target device is mounted on a PCB which facilitates to capture side-channel measurements

Oscilloscope (Attacker):

- ❑ Used to capture power consumption of the target device
- ❑ We can store the captured power measurements and send this information back to PC
- ❑ Two Channels:
 - ❑ Channel 1 (Yellow): Current/Power Measurements
 - ❑ Channel 2 (Blue): Trigger
- ❑ Why trigger is required??
- ❑ **Trigger**: Indication of when the encryption occurs!!!

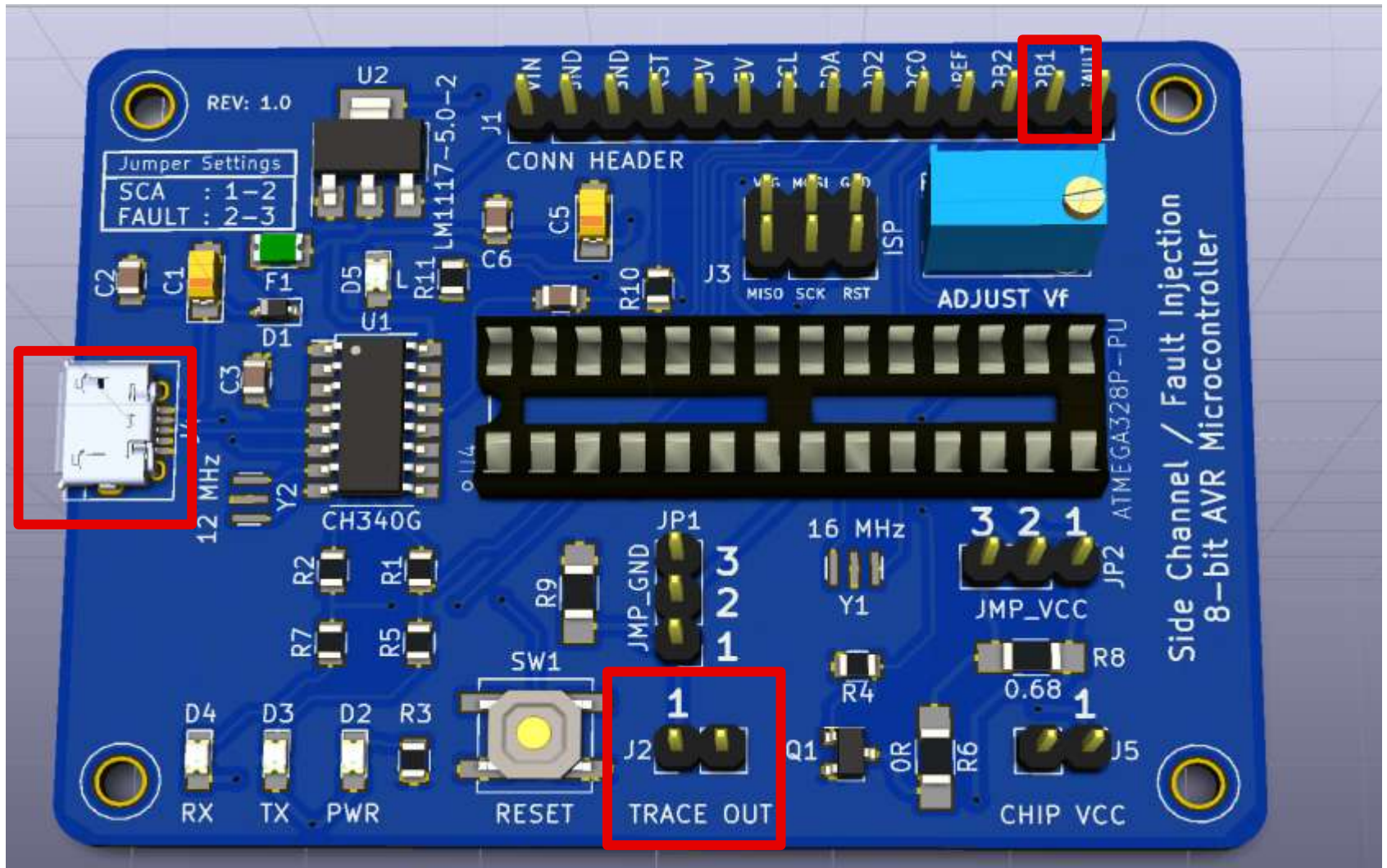
Experimental Setup:

Attacker



Target

Connect Target to Oscilloscope:



Power up the Board Using USB Cable (UART Header)

Python script to use

- ❑ Sca_lab_students.py
- ❑ Open Pycharm, and a new project in Conda environment (Python 3.8)
- ❑ Open Terminal to install some packages:
 - ❑ pip install matplotlib
 - ❑ pip install pyvisa
 - ❑ pip install pyserial
- ❑ Python script is good to go!!!

Step-1: Let us manually configure the scope

- ❑ Instrument: Rigol Oscilloscope DS1104

- ❑ Things to Configure:

 - ❑ Horizontal

 - ❑ Vertical

 - ❑ Trigger

Step-1: Let us manually configure the scope

- ❑ Instrument: Rigol Oscilloscope DS1104

- ❑ Things to Configure:

 - ❑ Turn on Channels 1 and 2

 - ❑ AC Coupling for Channel 1, DC Coupling for Channel 2

 - ❑ Set the voltage scale for trigger channel

 - ❑ Set trigger as Channel 2 (Set Edge Trigger, Set Trigger Level)

 - ❑ Set the horizontal scale

Steps to capture the trace

- ☐ Connect to Scope
- ☐ Configure the Scope
- ☐ Arm the scope (Wait for trigger)
- ☐ Communicate with Target (Target triggers the scope)
- ☐ Store the trace to the PC from the scope

Step-2: Let us automatically configure the scope

- ❑ Let us use the python script to automatically configure the scope
- ❑ Find out the values based on the manual configuration and fill it in the python script

How do devices leak “info” through side-channels?

- ❑ Data stored in memory and registers inside device
- ❑ When there is change in data in memory and registers, device consumes more current at that instance...
- ❑ When data switch does not happen, device consumes less current at that instance...

How do devices leak “info” through side-channels?

0	1	1	1	0	1	0	1
---	---	---	---	---	---	---	---

Switch



0	1	1	0	1	1	0	0
---	---	---	---	---	---	---	---

Power Consumption

B

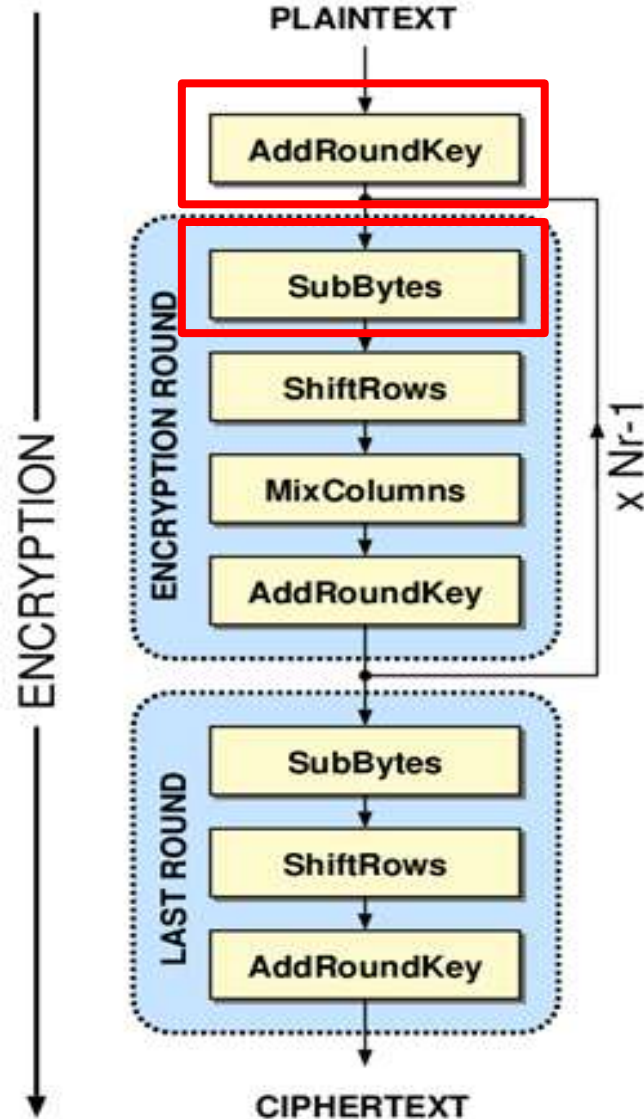
MORAL OF THE STORY:

MORE NUMBER OF BITS SWITCH, MORE CURRENT
LESS NUMBER OF BITS SWITCH, LESS CURRENT

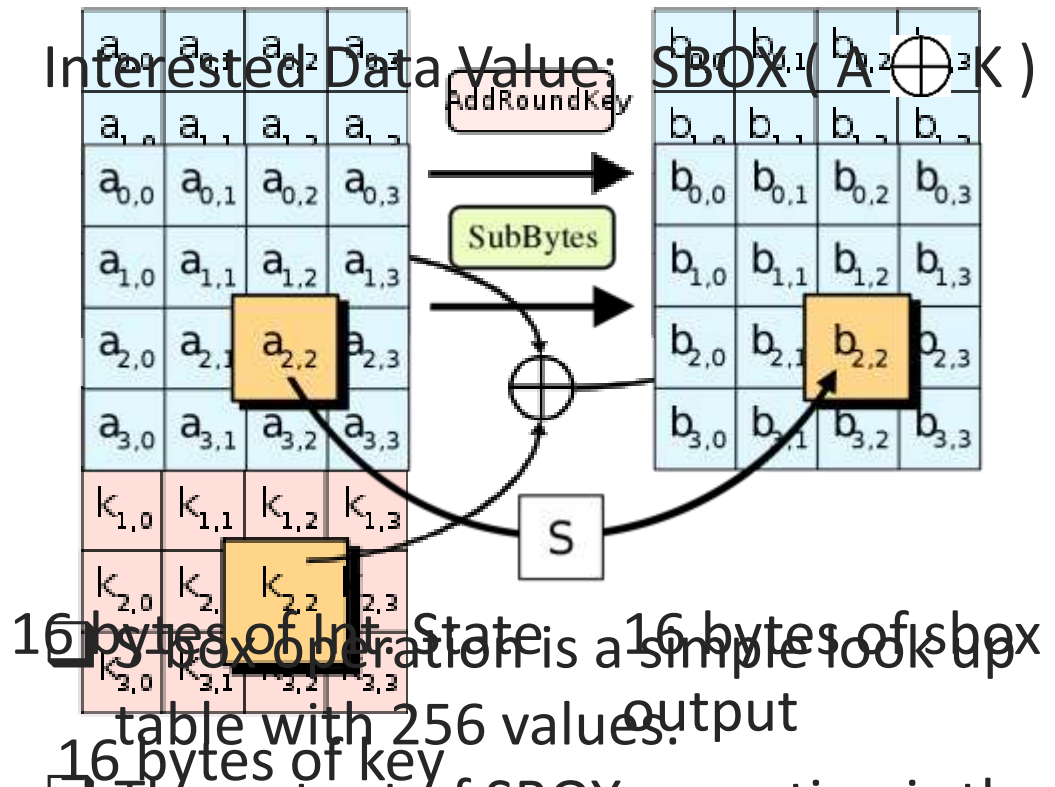
AES (Advanced Encryption Standard)

16 bytes of PT

16 bytes of Int. state



Interested Data Value:



- ❑ The output of SBOX operation is the intermediate value we use for our attack.

Power Analysis of AES

$$Y = \text{SBOX} (A \oplus K)$$

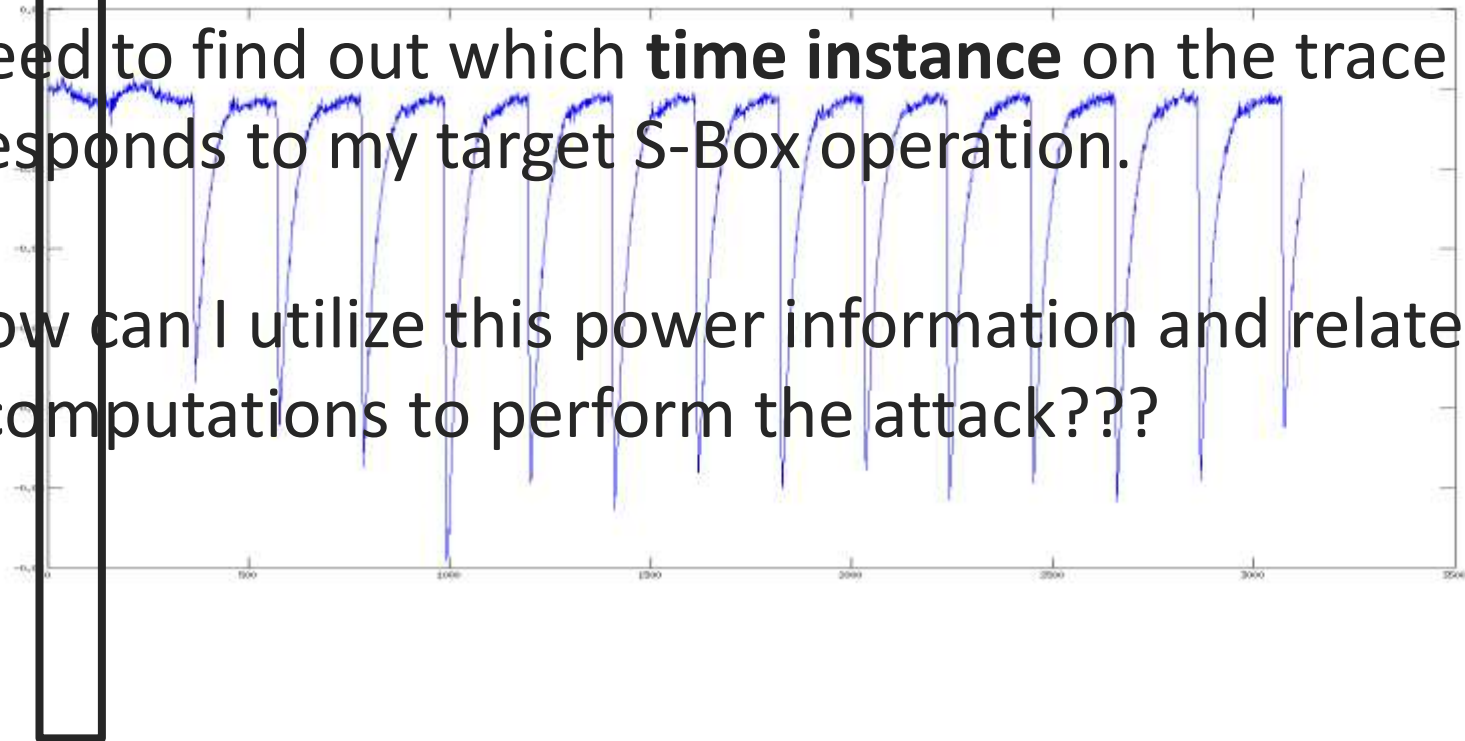
Target Value for attack

Natural Questions:

Where is my SBOX operation?

1) Need to find out which **time instance** on the trace corresponds to my target S-Box operation.

2) How can I utilize this power information and relate it to the computations to perform the attack???



Thank you